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# INVENTORY MANAGEMENT SYSTEM

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**MASTER OF COMPUTER APPLICATION**



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**GROUP No.: - 35**

## ACKNOWLEDGEMENT

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# 1) Introduction

## 1.1. Problem Definition

The Customer and Manager problem of time conflicts and communication gap. Inventory Management System can be design for the advanced time and which can be overcome time issue. It will also Reduce communication gap. And Manager Also Have All Details of All Product

## 1.2. Objective of the New System

The application can also generate the order reports to maintain the stock and order. For eg. generate the bill. This IMS portal is mainly use for mobile store. Which has functionalities of add product, update products, categories and product with stock management and also have functionality of billing which will have 3 status for paid, unpaid or partially payment.

## 1.3. Core Components

### **MANAGE CATEGORY:**

This IMS portal is mainly use for mobile store. Which has functionalities of add categories likes mobile phones and laptops and product with stock management.

## **MANAGE System:**

In system admin can manage all products.

ADD, UPDATE, DELETE and their other details.

## **MANAGE SERVICES DETAIL:**

Invoicing and billing application for different business purposes helps mainly the service providers and freelancers to manage, send professional invoices online, and track their status. Generally, all the small companies in various countries facing various issues for managing and tracking the invoice status of customers.

## **MANAGE INVOICE:**

This IMS is mainly use for mobile store. Which has functionalities of generating invoice.

## **GENERATE REPORT:**

Admin will be able to generate yearly, quarterly, monthly collection report as per the need.

## 1.4. Project Profile

|                    |                                        |                    |            |
|--------------------|----------------------------------------|--------------------|------------|
| PROJECT TITLE      | Inventory Management System            |                    |            |
| COMPONENTS/MODULES | Admin                                  |                    |            |
| INTERNAL GUIDE     | <b>Prof. VIDHI SAGAR</b>               |                    |            |
| DURATION           | 6 Months                               |                    |            |
| PLATFORM           | Web Application                        |                    |            |
| INSTITUTE NAME     | L.J. Institute Of Computer Application |                    |            |
| FRONT-END          | CSS,HTML,Bootstrap                     |                    |            |
| BACK-END           | MySQL, PHP                             |                    |            |
| DEVELOPMENT KIT    | PHP                                    |                    |            |
| SOFTWARES          | PHP                                    | Visual Studio Code | SQL Server |
|                    | CSS                                    | XAMPP Server       | HTML       |
|                    |                                        |                    |            |
| DEVELOPED BY       | MCA (SEM - 3)                          |                    |            |
|                    | <b>Ronak Sabhaya</b>                   | 2021004401110066   |            |
|                    | <b>Bhautik Parmar</b>                  | 2021002401110044   |            |
|                    | <b>Maaz Shaikh</b>                     | 2021004401110071   |            |

## 1.5. Advantages and Limitations of the Proposed System

### **ADVANTAGES:**

- All operations are accurate and user friendly. A modern approach to work in less time .
- Client gets all Details at home.
- Facilitate to maintain Stock and records.
- We get Proper Details on Inventory.
- Easy to maintain Product.

### **LIMITATIONS:**

- Always need an internet access for use this application.
- Stock has to be always updated.
- Updated Windows System

## 1.6. Proposed Time Line Chart

| Proposed Timeline Chart     |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
|-----------------------------|--------------|---|---|---|-----------|---|---|---|---------|----|----|----|----------|----|----|----|----------|----|
| Task                        | WEEKLY(2022) |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
|                             | August       |   |   |   | September |   |   |   | October |    |    |    | November |    |    |    | December |    |
|                             | 1            | 2 | 3 | 4 | 5         | 6 | 7 | 8 | 9       | 10 | 11 | 12 | 13       | 14 | 15 | 16 | 17       | 18 |
| Introduction                |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Problem Definition          |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Objective Of The New System |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Core Components             |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Project profile             |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Advantages And Limitation   |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Time Line Chart             |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Requirement Determination   |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Requirement Specification   |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Use Case Diagram            |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Class Diagram               |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Activity Diagram            |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Data Dictionary             |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| User Interface Design       |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Coding Standards            |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Test Case                   |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Proposed Enhancements       |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Conclusion                  |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Bibliography                |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Project Coding              |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |
| Project Documentation       |              |   |   |   |           |   |   |   |         |    |    |    |          |    |    |    |          |    |

## 2) Requirement Determination & Analysis

### 2.1. Requirement Determination

There are plethora of possible ways for requirement gathering like,

- Similar Application/Websites Study
- Questionnaires/Survey Other Manager
- Interaction with user (Client)

### 2.2. Requirement Specification

#### **FUNCTIONAL REQUIREMENTS:**

##### **ADMIN:**

##### **❖ Login**

- Login with Admin credential.

##### **❖ Manage Details**

- Can Add/Update/Delete Manager information from/to portal (System).
- 

#### **Constraints :**

##### **Hardware Constraints :**

- For better use of Visual Studio Code Ram must be of 2 or more GB as mentioned in Hardware Requirements. (For Developer)

## **Reliability Constraints**

- Validation is the main reliability requirement that is used in the system. Without proper validation, system does not allow to login in the system.

## **PERFORMANCE REQUIREMENTS:**

- ❖ The system must be able to perform smoothly without any interruptions.
- ❖ There should be minimum interruption in connectivity.

## **SECURITY REQUIREMENTS:**

- ❖ All the in the database should have backup in case of data loss.

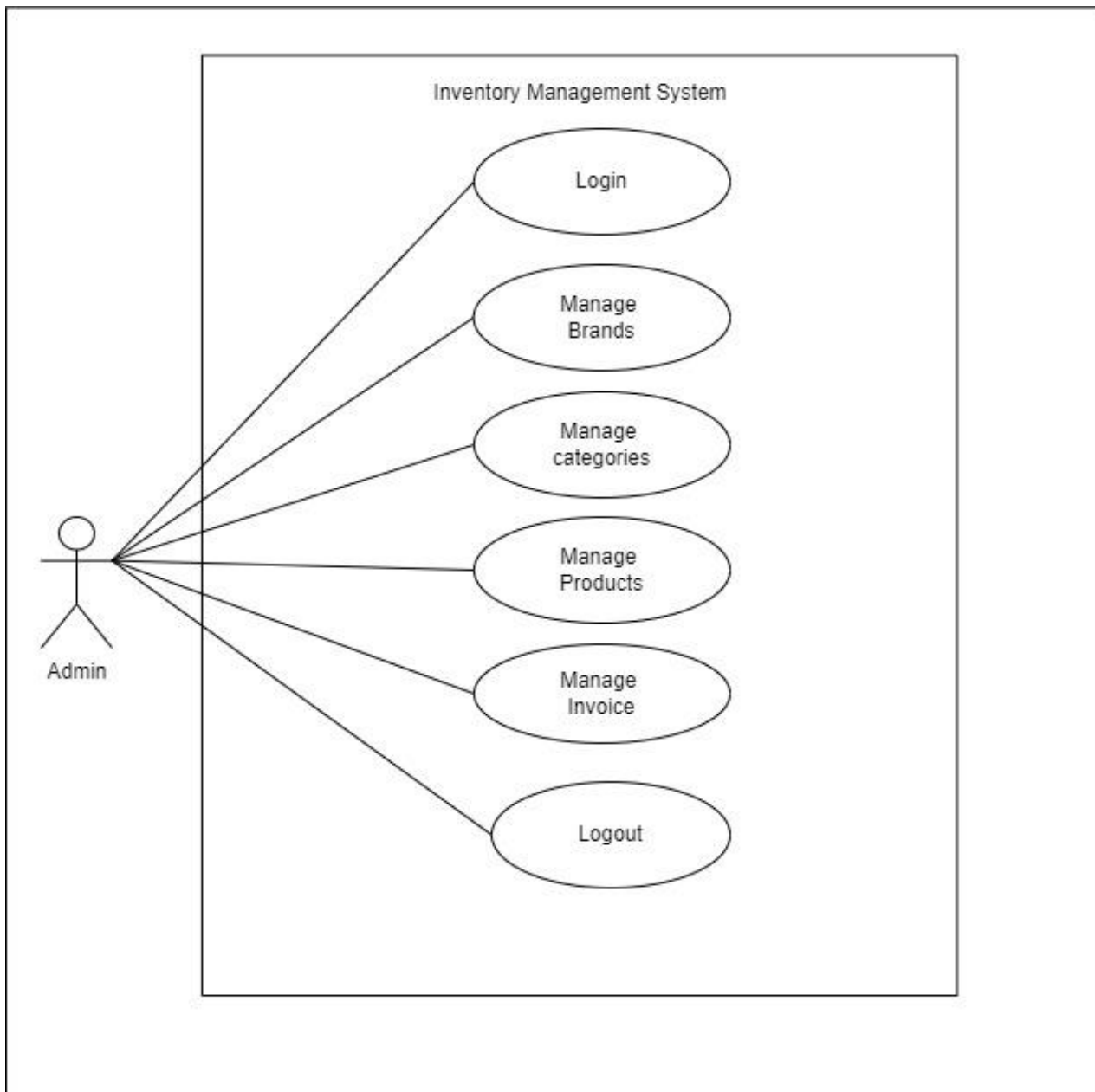
## **\* HARDWARES REQUIREMENTS:**

### **HARDWARES:**

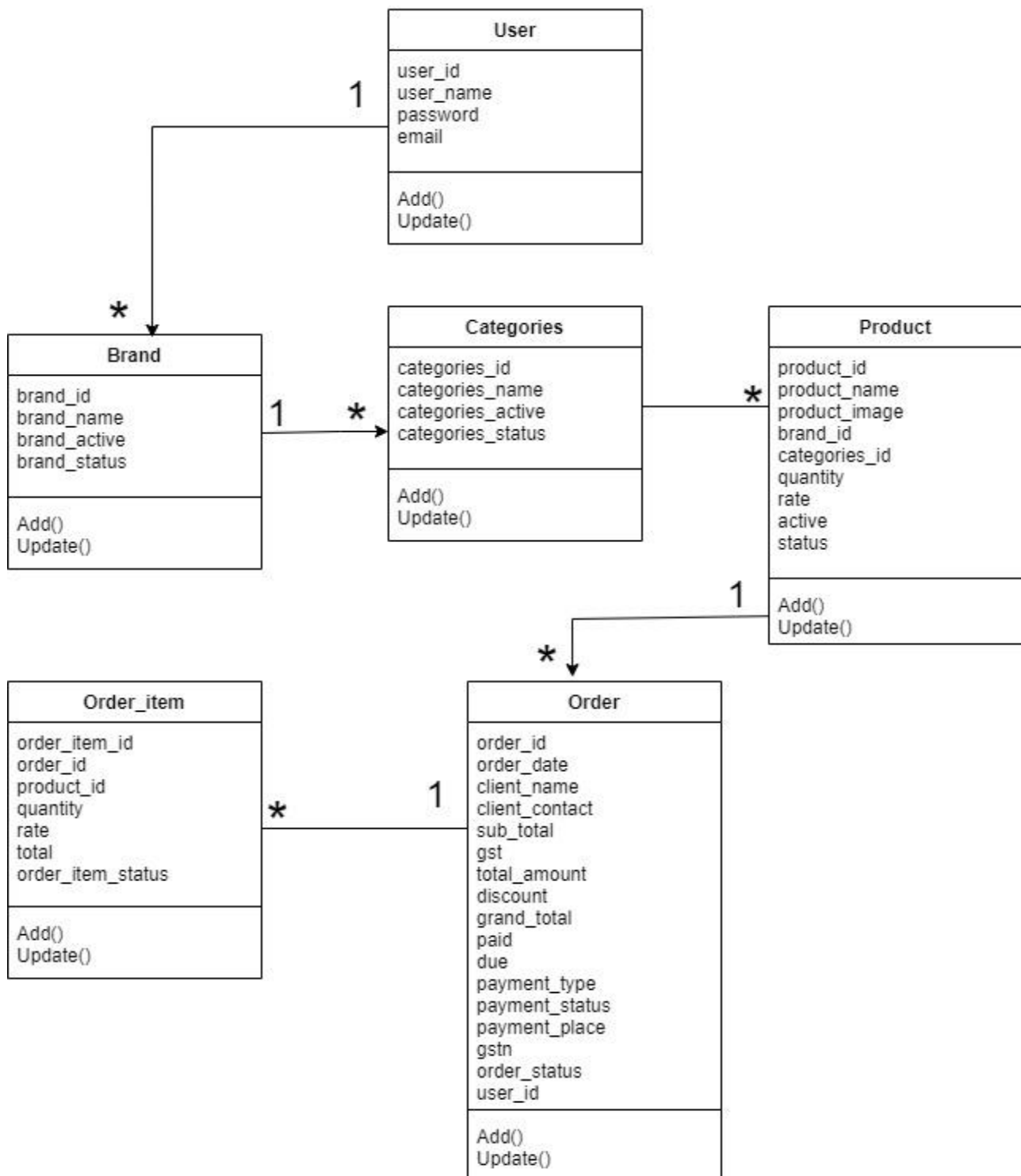
- ❖ Windows PC & Laptop
  - Windows Version up to Win7

### 3) System Design

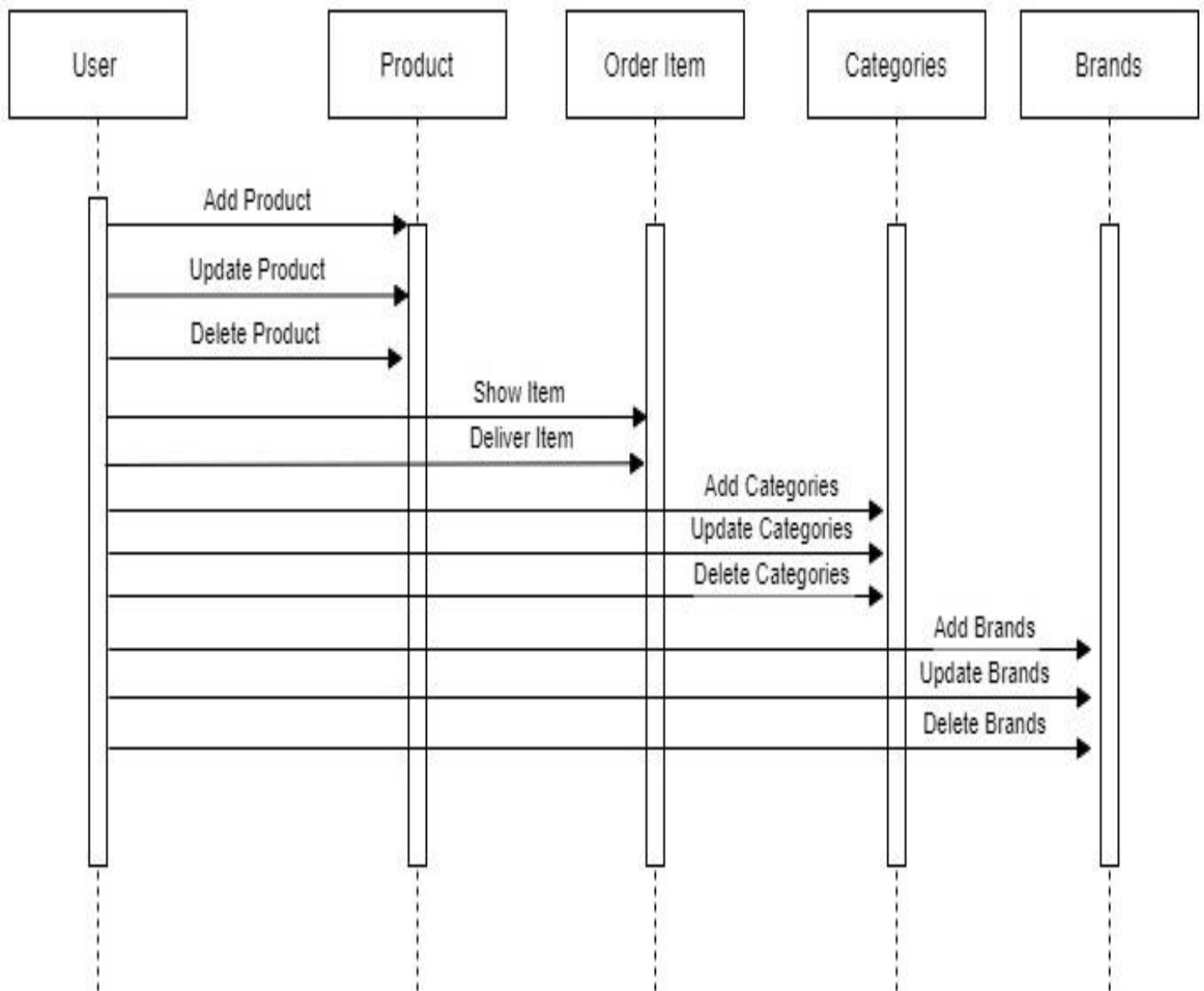
#### 3.1. Use Case Diagram



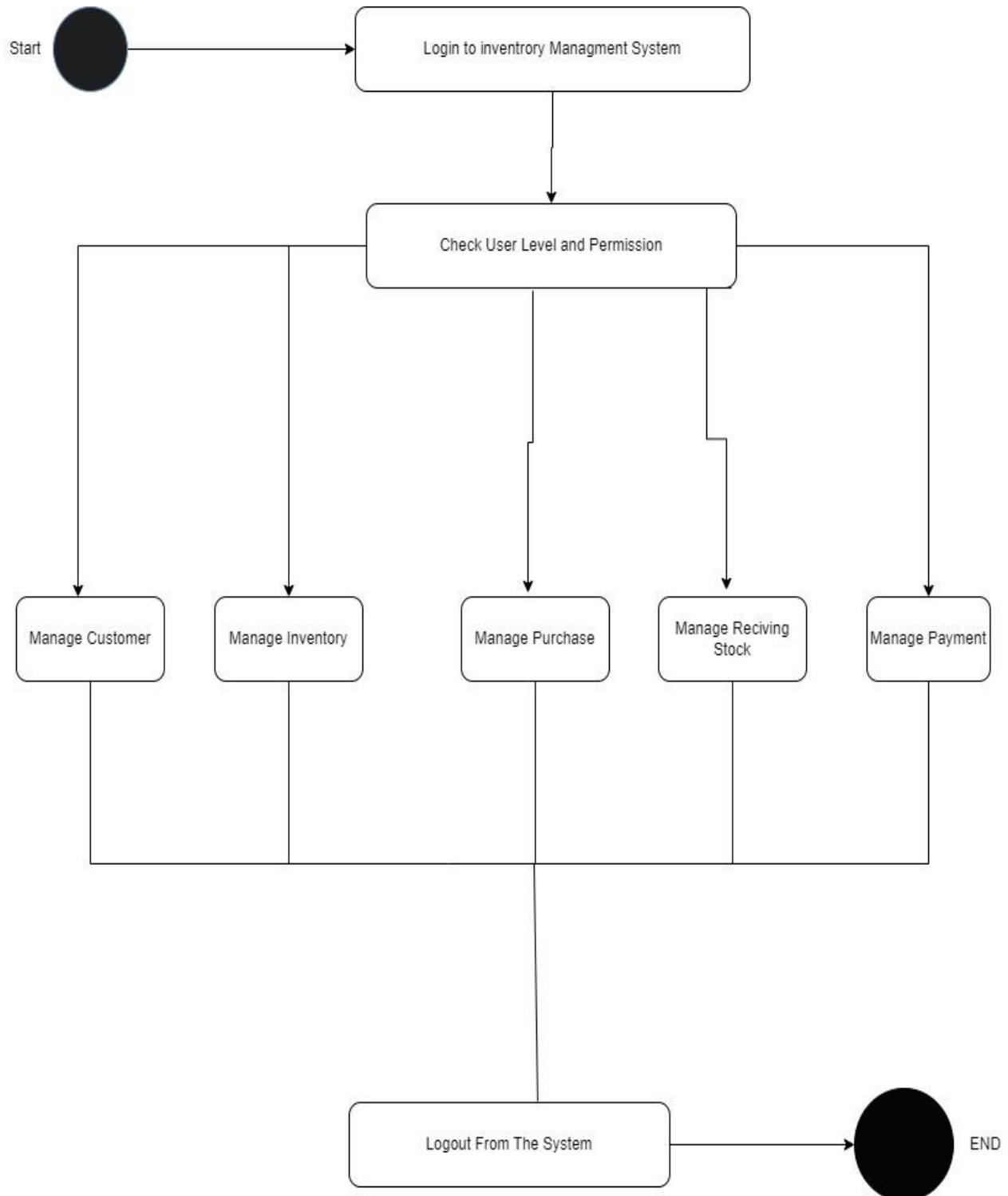
## 3.2. Class Diagram



### 3.3. Interaction Diagram



### 3.4. Activity Diagram



### 3.5. Data Dictionary

#### User Login

| Sr.No | Field name | Data type | size | Constraint  | Description                    |
|-------|------------|-----------|------|-------------|--------------------------------|
| 1     | user_id    | Int       | 11   | Primary Key | This field store user login id |
| 2     | user_name  | Varchar   | 25   | Not Null    | This field store user name     |
| 3     | Password   | Varchar   | 25   | Not Null    | This field store user password |
| 4     | Email      | Varchar   | 25   | Not Null    | This field store user email    |

#### Brand

| Sr.No | Field name   | Data type | Size | Constraint  | Description                   |
|-------|--------------|-----------|------|-------------|-------------------------------|
| 1     | Brand_Id     | Int       | 11   | Primary Key | This field store id           |
| 2     | Brand_name   | Varchar   | 25   | Not Null    | This field store brand_name   |
| 3     | Brand_active | Int       | 11   | Not Null    | This field store brand_active |
| 4     | Brand_status | Int       | 11   | Not Null    | This field store brand_status |

## Categories

| Sr.No | Field name        | Data type | Size | Constraint  | Description                        |
|-------|-------------------|-----------|------|-------------|------------------------------------|
| 1     | Categories_Id     | Int       | 11   | Primary Key | This field store id                |
| 2     | Categories_name   | Varchar   | 25   | Not Null    | This field store Categories_name   |
| 3     | Categories_active | Int       | 11   | Not Null    | This field store Categories_active |
| 4     | Categories_status | Int       | 11   | Not Null    | This field store Categories_status |

## Product

| Sr No | Field name    | Data type | Size | Constraint  | Description                    |
|-------|---------------|-----------|------|-------------|--------------------------------|
| 1     | product_id    | Int       | 11   | Primary Key | This field store product_id    |
| 2     | product_name  | Varchar   | 25   | Not Null    | This field store product_name  |
| 3     | product_image | Text      | -    | Not Null    | This field store product_image |
| 4     | brand_id      | Int       | 11   | Foreign Key | This field store brand_id      |
| 5     | categories_id | int       | 11   | Not Null    | This field store categories_id |
| 6     | quantity      | Varchar   | 25   | Not Null    | This field store quantity      |
| 7     | Rate          | Varchar   | 25   | Not Null    | This field store rate          |
| 8     | active        | Int       | 11   | Not Null    | This field store active        |
| 9     | Status        | Int       | 11   | Not Null    | This field store status        |

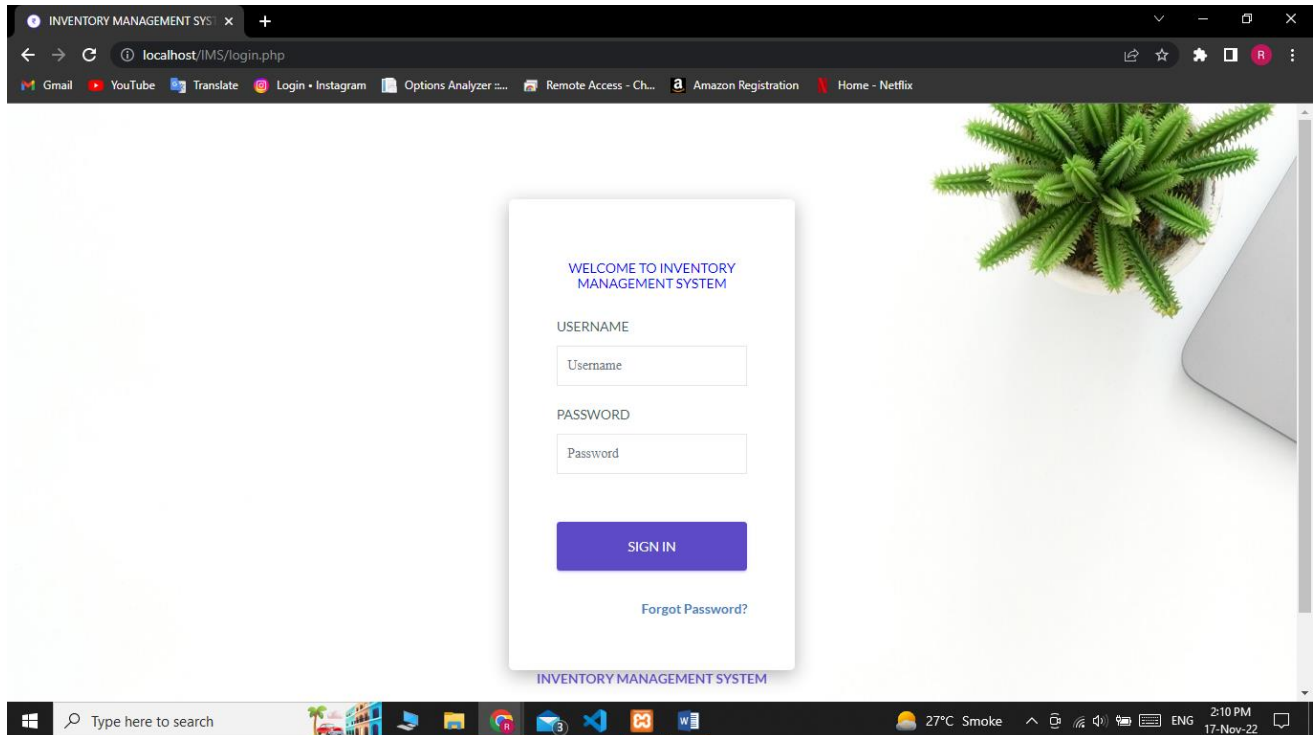
## Orders

| Sr.No | Field name     | Data type | Size | Constraint  | Description                     |
|-------|----------------|-----------|------|-------------|---------------------------------|
| 1     | Order_Id       | Int       | 11   | Primary Key | This field store id             |
| 2     | Order_date     | Date      | -    | Not Null    | This field store order_date     |
| 3     | client_name    | Varchar   | 25   | Not Null    | This field store client_name    |
| 4     | client_contact | Varchar   | 25   | Not Null    | This field store client_contact |
| 5     | sub_total      | Varchar   | 25   | Not Null    | This field store sub_total      |
| 6     | Vat            | Varchar   | 25   | Not Null    | This field store vat            |
| 7     | total_amount   | Varchar   | 25   | Not Null    | This field store total_amount   |
| 8     | Discount       | Varchar   | 25   |             | This field store Discount       |
| 9     | grand_total    | Varchar   | 25   | Not Null    | This field store grand_total    |
| 10    | Paid           | Varchar   | 25   | Not Null    | This field store Paid           |
| 11    | Due            | Varchar   | 25   | Not Null    | This field store Due            |
| 12    | payment_type   | Int       | 11   | Not Null    | This field store payment_type   |
| 13    | payment_status | Int       | 11   | Not Null    | This field store payment_status |
| 14    | payment_place  | Int       | 11   | Not Null    | This field store payment_place  |
| 15    | Gstn           | Varchar   | 25   | Not Null    | This field store Gstn           |
| 16    | order_status   | Int       | 11   | Not Null    | This field store order_status   |
| 17    | user_id        | Int       | 11   | Not Null    | This field store user_id        |

## Order\_Item

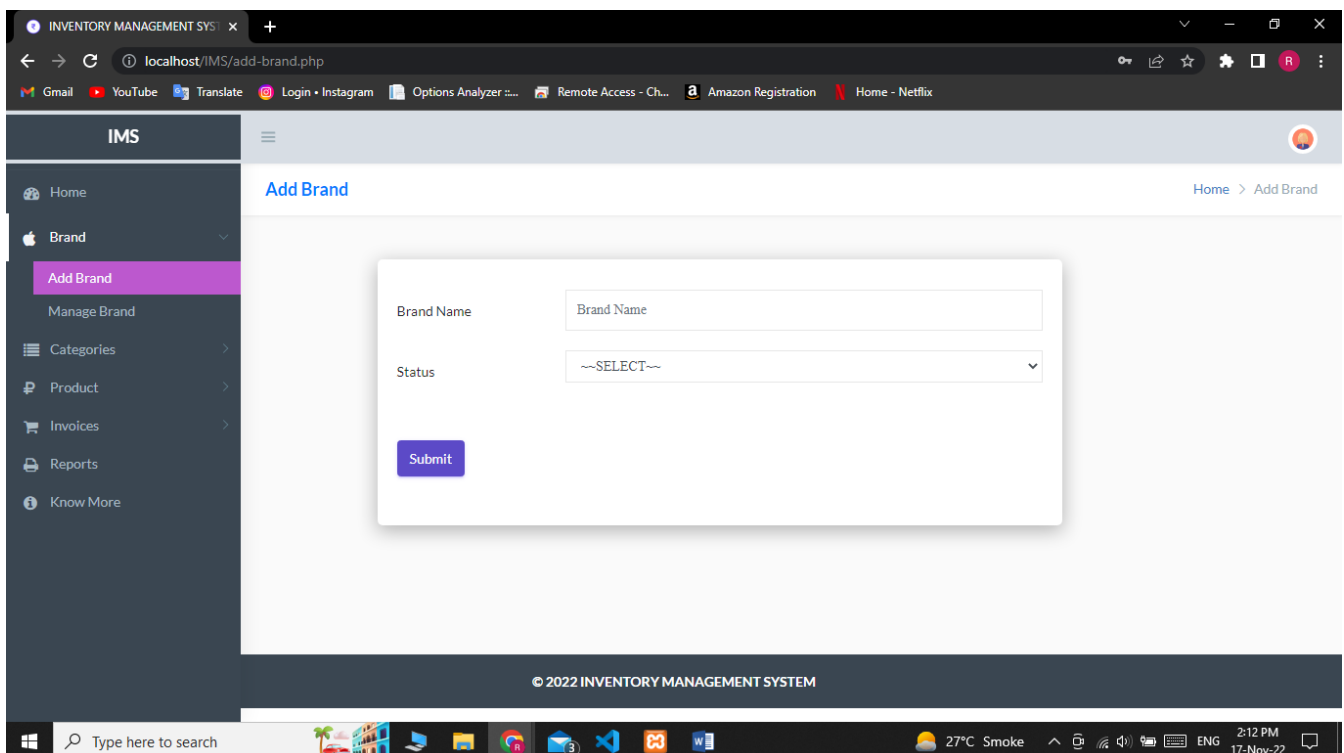
| Sr.No | Field name        | Data type | Size | Constraint  | Description                        |
|-------|-------------------|-----------|------|-------------|------------------------------------|
| 1     | order_item_id     | Int       | 11   | Primary Key | This field store order_item_id     |
| 2     | order_id          | Int       | 11   | Foreign Key | This field store order_id          |
| 3     | product_id        | Int       | 11   | Foreign Key | This field store product_id        |
| 4     | quantity          | Varchar   | 25   | Not Null    | This field store quantity          |
| 5     | Rate              | Varchar   | 25   | Not Null    | This field store Rate              |
| 6     | Total             | Varchar   | 25   | Not Null    | This field store Total             |
| 7     | order_item_status | int       | 11   | Not Null    | This field store order_item_status |

## 3.6. User Interface Design



The screenshot shows a web browser window with the URL `localhost/IMS/login.php`. The page features a login form titled "WELCOME TO INVENTORY MANAGEMENT SYSTEM". The form includes fields for "USERNAME" and "PASSWORD", a "SIGN IN" button, and a "Forgot Password?" link. The background of the page shows a desk with a laptop and a potted plant. The Windows taskbar at the bottom displays the time as 2:10 PM on 17-Nov-22.

INVENTORY MANAGEMENT SYSTEM



The screenshot shows the "Add Brand" page of the Inventory Management System. The page has a sidebar menu with options: Home, Brand, Add Brand (selected), Manage Brand, Categories, Product, Invoices, Reports, and Know More. The main content area contains a form with a "Brand Name" input field and a "Status" dropdown menu with a "Submit" button. The footer of the page reads "© 2022 INVENTORY MANAGEMENT SYSTEM". The Windows taskbar at the bottom displays the time as 2:12 PM on 17-Nov-22.

IMS

Add Brand

Home > Add Brand

Brand Name

Status

Submit

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INVENTORY MANAGEMENT SYS

localhost/IMS/brand.php

Home > View Brand

Add Brand

Show 10 entries

| # | Brand Name | Status    | Action                                      |
|---|------------|-----------|---------------------------------------------|
| 1 | Onewplus   | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2 | Apple      | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3 | Samsung    | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4 | HP         | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 5 | Apple      | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 6 | MI         | Available | <a href="#">Edit</a> <a href="#">Delete</a> |

Showing 1 to 1 of 1 entries

Previous 1 Next

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INVENTORY MANAGEMENT SYS

localhost/IMS/add-category.php

Home > Add Categories Management

Add Category

Categories Name

Status

Submit

© 2022 INVENTORY MANAGEMENT SYSTEM

INVENTORY MANAGEMENT SYSTEM

localhost/IMS/categories.php

Home > View Categories

**View Categories**

[Add Categories](#)

Show 10 entries

| # | Categories Name | Status    | Action                                      |
|---|-----------------|-----------|---------------------------------------------|
| 1 | Android         | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2 | MacOs           | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3 | Window          | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4 | Tab             | Available | <a href="#">Edit</a> <a href="#">Delete</a> |

Showing 1 to 1 of 1 entries

Previous 1 Next

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INVENTORY MANAGEMENT SYSTEM

localhost/IMS/add-product.php

Home > Add Product

**Add Product**

Product Image: [Choose File](#) No file chosen

Product Name:

Quantity:

Rate:

Brand Name:

Category Name:

Status:

[Submit](#)

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INVENTORY MANAGEMENT SYSTEM

localhost/IMS/product.php

Home > View Product

[Add Product](#)

Show 10 entries

| # | Photo                                                                             | Product Name  | Rate  | Quantity | Brand    | Category | Status    | Action                                      |
|---|-----------------------------------------------------------------------------------|---------------|-------|----------|----------|----------|-----------|---------------------------------------------|
| 1 |  | Onesplus 9R   | 43999 | 11       | Onesplus | Android  | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2 |  | HP Pavilion   | 48999 | 3        | HP       | Window   | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3 |  | iPhone 13 Pro | 71900 | 7        | Apple    | MacOs    | Available | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4 |  | MI 12         | 15999 | 4        | MI       | Android  | Available | <a href="#">Edit</a> <a href="#">Delete</a> |

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INVENTORY MANAGEMENT SYSTEM

localhost/IMS/add-order.php

Home > Create Invoice

[Add Invoice](#)

Invoice No:  Invoice Date:

Client Name:  Client Contact No.:

| Product  | Rate | Avail. | Quantity | Total | Action                                     |
|----------|------|--------|----------|-------|--------------------------------------------|
| ~SELECT~ |      |        |          |       | <a href="#">Add</a> <a href="#">Delete</a> |

Sub Amount:  Total Amount:

Discount:  Grand Total:

GST 18%:  Paid Amount:

Due Amount:  Payment Type:

Payment Status:  Payment Place:

[Submit](#) [Reset](#)

© 2022 INVENTORY MANAGEMENT SYSTEM

INVENTORY MANAGEMENT SYS x +

localhost/IMS/Order.php

Gmail YouTube Translate Login • Instagram Options Analyzer ... Remote Access - Ch... Amazon Registration Home - Netflix

IMS

Home

Brand

Categories

Product

Invoices

Add Invoice

Manage Invoices

Reports

Know More

View Invoice

Home > View Invoice

Add Invoice

Show 10 entries

Search:

| # | Invoice Date | Client Name   | Contact    | Total Invoice Item | Payment Status  | Action                                      |
|---|--------------|---------------|------------|--------------------|-----------------|---------------------------------------------|
| 1 | 2022-09-26   | Ronak Sabhaya | 1234567890 |                    | Advance Payment | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2 | 2022-11-11   | Ronak Patel   | 1597536842 |                    | Advance Payment | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3 | 2022-11-15   | Ronak         | 8156052038 |                    | Full Payment    | <a href="#">Edit</a> <a href="#">Delete</a> |

Showing 1 to 1 of 1 entries

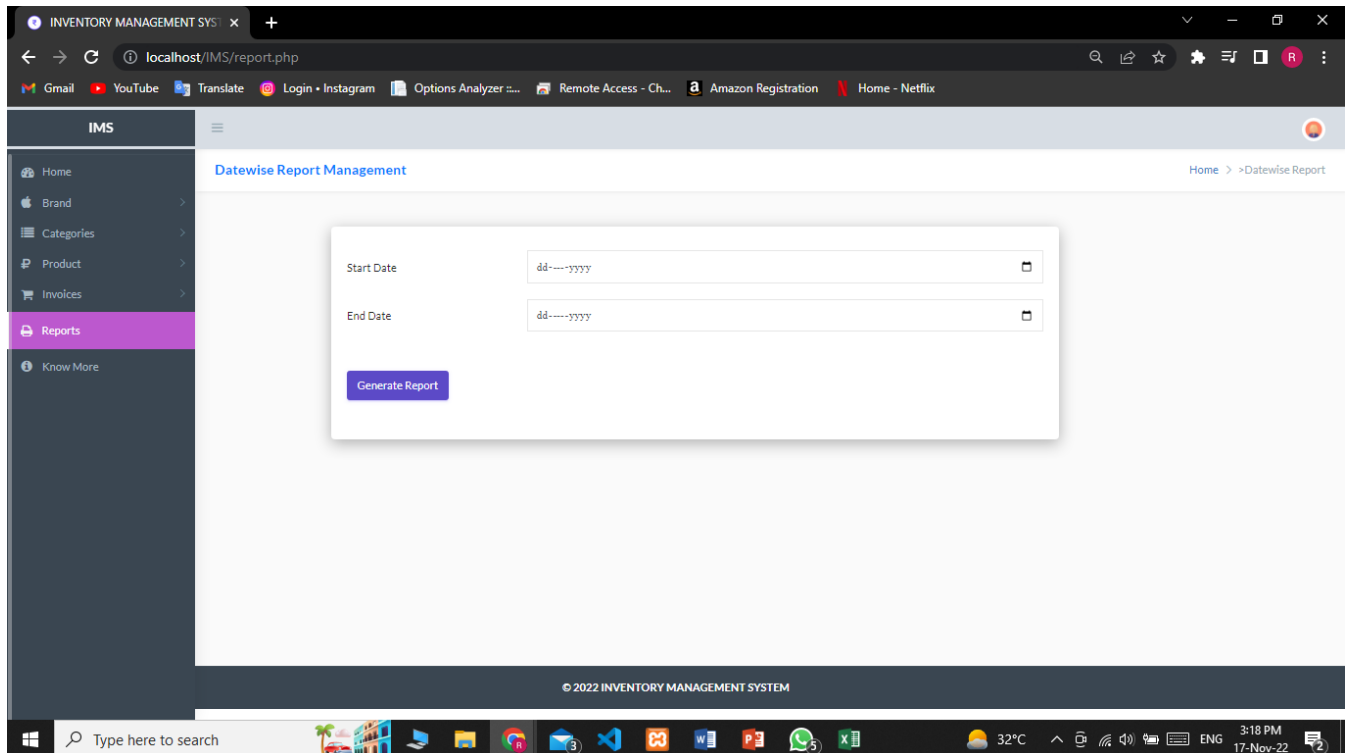
Previous 1 Next

© 2022 INVENTORY MANAGEMENT SYSTEM

Type here to search

32°C 3:17 PM 17-Nov-22

## 3.7 Report Design



INVENTORY MANAGEMENT SYS

localhost/IMS/report.php

IMS

Home > Datewise Report

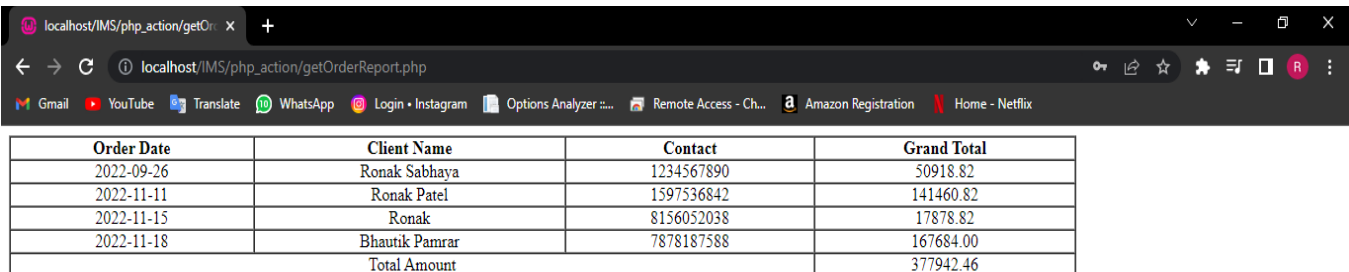
Datewise Report Management

Start Date: dd-yyy

End Date: dd-yyy

Generate Report

© 2022 INVENTORY MANAGEMENT SYSTEM



| Order Date   | Client Name    | Contact    | Grand Total |
|--------------|----------------|------------|-------------|
| 2022-09-26   | Ronak Sabhaya  | 1234567890 | 50918.82    |
| 2022-11-11   | Ronak Patel    | 1597536842 | 141460.82   |
| 2022-11-15   | Ronak          | 8156052038 | 17878.82    |
| 2022-11-18   | Bhautik Pamrar | 7878187588 | 167684.00   |
| Total Amount |                |            | 377942.46   |

## 4) Development

### 4.1. Coding Standard

- ❖ Class and method names should always have be in proper format.
- ❖ Method arguments and Local variable should always in Camel case.
- ❖ Avoid the use of underscore while naming identifier.
- ❖ Use proper alignment for coding.

## 5) Test Cases

| <u>Test Case</u> | <u>Test Case Description</u>                    | <u>Test Case Input</u>                      | <u>Test Data</u>                                 | <u>Post Condition</u>                   | <u>Expected Output</u> | <u>Actual Result</u> |
|------------------|-------------------------------------------------|---------------------------------------------|--------------------------------------------------|-----------------------------------------|------------------------|----------------------|
| <u>1</u>         | <u>Verify the login Funcality Of Login Page</u> | <u>Enter valid User Name &amp; Password</u> | <u>User: admin</u><br><u>Password: rootadmin</u> | <u>Success massage</u>                  | <u>Success</u>         | <u>Valid</u>         |
| <u>2</u>         | <u>Verify the login Funcality Of Login Page</u> | <u>Enter valid User Name &amp; Password</u> | <u>User: Admin</u><br><u>Password: Rootadmin</u> | <u>Invalid user name &amp; password</u> | <u>Error</u>           | <u>Invalid</u>       |

## 6) Proposed Enhancements

- ❖ Add a feature of reminder of payment .
- ❖ Make a centralize inventory management system for warehouse.

## 7) Conclusion

Inventory Management System has been completed successfully.

The System is operated at a high level of efficiency and all the Admin with the system understands its advantages.

We have tried my level best to make this portal as dynamic as possible. Also, provision is provided for future development in the Inventory Management System.

Inventory Management System with portal will be approved and implemented soon

## 8) Bibliography

- URL: [www.getbootstrap.com](http://www.getbootstrap.com)
- URL: [www.w3schools.com](http://www.w3schools.com)
- URL: [www.youtube.com](http://www.youtube.com)